Zezhong Qian

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Research Interests

World Models • Embodied AI • Diffusion Models • Vision-Language-Action • Robotics Reinforcement Learning

Education

Peking University 2027 – 2032
Ph.D. in Computer Science, School of Computer Science
Advisor: Prof. Shanghang Zhang

Xi'an Jiaotong University 2023 – 2027
B.E. in Artificial Intelligence
GPA: 93.59/100 Ranked 2/66

Young Gifted Program Preparatory, Xi'an Jiaotong University 2022 – 2023
(Accelerated Program)

Research Papers

DualDiff: Dual-branch Diffusion Model for Autonomous Driving with Semantic Fusion (ICRA 2025) 2024
Third Author Generative World Model for Autonomous Driving

- Identified geometric inconsistency under viewpoint perturbation in diffusion-based driving world models.
- Proposed **Occupancy Ray Sampling**, enforcing cross-view structural alignment during generative sampling.
- Improved multi-view fidelity and reduced scene distortion in long-horizon rollouts.

GRID-Planner: Graph-Structured Relational Reward Shaping for Diffusion Planning(ACMMM 2026) 2025
Co-first Author On-policy RL for Flow-Matching Planner

- Tackled long-tail safety failures in diffusion-based driving planners.
- Built the RL module with relational rewards, advantage-weighted updates, and on-policy distillation.
- Improved planning score by 3.4%, reducing collision, lane deviation, and jerk by 35.8%, 45.1%, and 21.2%.

CityGen: Structure-Guided City-Style Synthesis for Cross-City Autonomous Driving (ICME 2026 Spotlight) 2025
First Author Cross-domain World Modeling

- Investigated domain shift in generative driving simulators across different city layouts.
- Designed a structure-guided synthesis module to disentangle geometry from style.
- Improved cross-city generalization and reduced structural artifacts in generated scenes.

WristWorld: Generating Wrist-Views via 4D World Models for Robotic Manipulation (NeurIPS Under Review) 2025
First Author Egocentric 4D World Model for Robotics

- Identified the limitation of third-person 4D world models in egocentric robotic manipulation.
- Proposed a geometry-constrained wrist-view world model enforcing temporal-view consistency.
- Introduced a cross-view latent alignment module to support embodied interaction.

WA-RL: World-Action Model Reinforcement Learning with Reconstruction Rewards and Online Video SFT (CVPR Workshop 2026) 2026
First Author World-Action Reinforcement Learning for Robotics

- Tackled demonstration-limited learning in World-Action robotic models.
- Built WA-RL to jointly optimize the actor and world model with reconstruction rewards and online video SFT.

- Improved LIBERO and RLBench performance, showing stronger recovery and long-horizon manipulation.

Hierarchical Denoising for Multi-Step Visual Reasoning (NeurIPS 2026 Under Review) 2026

First Author Video Reasoning

- Addressed the core tension in video reasoning: streaming autoregressive diffusion(e.g. CausVid, CausalForcing) enables low-latency generation but struggles with multi-step logical consistency, while bidirectional diffusion supports global revision at high inference cost.
- Proposed HDR, a hierarchical denoising framework that organizes video latents into a tree-structured hierarchy for coarse-to-fine reasoning, with sparse hierarchical attention to preserve streaming efficiency.
- Improved overall success from 34.22 to 60.29 and average progress from 76.00 to 89.56 across six multi-step reasoning tasks, while achieving 0.70s per latent streaming latency and strong data efficiency.

Research Experience

Beijing Humanoid Innovation Center 2026 – Present

Research Intern

- Achieved **10× inference speedup** via DMD-based distillation based on Causal Forcing.
- Converted open-loop generative model into causal closed-loop architecture.
- Conducted Vision-Action RL experiments for real-world humanoid deployment.

HMI Lab 2025 – Present

Research Intern (Mentor: Shanghang Zhang)

- Trained a **1.3B-parameter world model** on 64-GPU cluster.
- Led WristWorld project and authored view-transfer module.
- Contributed to technical report: *WoW: Towards a World-Omniscient World Model*.

Cytoderm 2024 – 2025

Research Intern

- Developed Video-Prediction-Policy (VPP) variants with history-conditioned prediction.
- Improved manipulation success rate on deformable object tasks.
- Contributed to RL-based control for desktop robot system.

IAIR, Xi'an Jiaotong University 2023 – 2025

Research Intern (Advisor: Longjun Liu)

- Researched generative world models for autonomous driving.
- Co-developed diffusion-based driving simulation engines.

Technical Skills

Programming Languages: Python (primary), C++
AI Engineering Tools: Codex, Cursor
Deep Learning Frameworks: PyTorch, TensorFlow
Large-Scale Training & Systems: DeepSpeed, Fully Sharded Data Parallel (FSDP), multi-node distributed training (64+ GPU clusters)
Robotics & Simulation: ROS, LeRobot
Research Areas: Diffusion Models, World Models, Vision-Language-Action (VLA), Reinforcement Learning, Embodied AI

Awards

National Scholarship (Top Academic Award)
National First Prize, RoboMaster Mech Master Super Competition (2024)
Gold Medal, ICPC Shaanxi Provincial Contest (2024)
Provincial First Prize, CUMCM (2024)